

Exercise 1

Write the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \|x\|_p \\ \text{subj. to} \quad & Ax \leq b \end{aligned}$$

as Linear Program (LP) for

- i) $p = 1$
- ii) $p = \infty$

Solution:

i) $p = 1$ (Slide 31 Lec.4):

- $\|x\|_1 = \sum_{i=1}^n |x_i|$
- Let $\epsilon \in \mathbb{R}^n$
- If $x \leq \epsilon$ and $-x \leq \epsilon$ it holds $|x_i| \leq \epsilon_i$
- Therefore $\|x\|_1 = \sum_{i=1}^n |x_i| \leq \sum_{i=1}^n \epsilon_i = 1_n^\top \epsilon$
- Equivalent formulation

$$\begin{aligned} \min_{x, \epsilon \in \mathbb{R}^n} \quad & 1_n^\top \epsilon \\ \text{subj. to} \quad & Ax \leq b \\ & x \leq \epsilon \\ & -x \leq \epsilon, \end{aligned}$$

since the optimal solution $\epsilon_i^* = |x_i|$ for which $1_n^\top \epsilon^* = \|x\|_1$ always beats any feasible $\tilde{\epsilon}$ since $1_n^\top \tilde{\epsilon} \geq 1_n^\top \epsilon^*$.

ii) $p = \infty$ (Slide 29 Lec.4):

- $\|x\|_\infty = \max_i |x_i|$
- Let $\epsilon \in \mathbb{R}$
- If $x \leq 1_n \epsilon$ and $-x \leq 1_n \epsilon$ it holds $|x_i| \leq \epsilon$ and therefore $\|x\|_\infty \leq \epsilon$.
- Similar argument leads to equivalent formulation

$$\begin{aligned} \min_{x \in \mathbb{R}^n, \epsilon \in \mathbb{R}} \quad & \epsilon \\ \text{subj. to} \quad & Ax \leq b \\ & x \leq 1_n \epsilon \\ & -x \leq 1_n \epsilon. \end{aligned}$$

Exercise 2

Write the optimization

$$\begin{aligned}
 J^*(x(k)) = \min_{x_i, u_i} & x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i \\
 \text{s.t. } & x_{i+1} = A x_i + B u_i, \quad \forall i = 0, \dots, N-1 \\
 & G_x x_i \leq b_x, \quad \forall i = 1, \dots, N-1 \\
 & G_u u_i \leq b_u, \quad \forall i = 0, \dots, N-1 \\
 & G_f x_N \leq b_f, \\
 & x_0 = x(k)
 \end{aligned}$$

as a QP with

$$\begin{aligned}
 J^*(x(k)) = \min_z & z^T H z + h^T z + c \\
 \text{s.t. } & G z \leq w \\
 & G_{eq} z = E_{eq} x(k)
 \end{aligned}$$

Solution:

Define: $z = [x_1^T \ x_2^T \ \dots \ x_N^T \ u_0^T \ u_1^T \ \dots \ u_{N-1}^T]^T$

The system dynamics $x_1 = A x_0 + B u_0$, $x_2 = A x_1 + B u_1$, ..., $x_N = A x_{N-1} + B u_{N-1}$ in matrix form yields

$$\underbrace{\begin{bmatrix} I & & & & -B & & \\ -A & I & & & -B & & \\ & -A & I & & -B & & \\ & & \ddots & \ddots & & \ddots & \\ & & & -A & I & & \\ & & & & & & -B \end{bmatrix}}_{G_{eq}} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \\ u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix} = \underbrace{\begin{bmatrix} A \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{E_{eq}} x(k),$$

the constraints in matrix form yield

$$\underbrace{\begin{bmatrix} G_x & & & & \\ & \ddots & & & \\ & & G_x & & \\ & & & G_f & \\ & & & & G_u & \ddots \\ & & & & & G_u \end{bmatrix}}_G \begin{bmatrix} x_1 \\ \vdots \\ x_{N-1} \\ x_N \\ u_0 \\ \vdots \\ u_{N-1} \end{bmatrix} \leq \underbrace{\begin{bmatrix} b_x \\ \vdots \\ b_x \\ b_f \\ b_u \\ \vdots \\ b_u \end{bmatrix}}_w,$$

and the cost matrices are

$$H = \begin{bmatrix} Q & & & & \\ & \ddots & & & \\ & & Q & & \\ & & & P & \\ & & & & R \\ & & & & & \ddots \\ & & & & & & R \end{bmatrix}, h = 0, c = x(k)^\top Q x(k).$$

The unspecified matrix entries are zero.